Project Sentinel · Official Submission Manifest

Cryptographic Package Verification & Component Functional Index

Kempower Challenge · LahtiES Energy Tech Hackathon 2026 (LUT University Lahti / CrazyTown)

- **Package Archive**: `Team3_Project_Sentinel_Submission.zip` (also accessible as `submission.zip`)
- **Total Deliverable Files**: 178 (including this manifest & its companion PDF)
- **Total Uncompressed Volume**: 28.61 MB (30,000,036 bytes)
- **Zero-CJK Rule Compliance**: 100% Pure Professional English Verified (`[\u4e00-\u9fff]` match count = 0)
- **Deterministic Verification**: `python 2_The_Demonstration_Part2/verify_all.py` (29 checks passing in < 0.5s)
- **Automated Test Coverage**: 462 Pytest + 106 E2E = 568 total automated tests (100% pass rate)

Deliverable Summary by Pillar

- `root/` (Top-level documentation & manifest): 2 files (607.67 KB)
- `1_pitch_and_story_2029/` (Saturday Deliverable 1: 10-Minute Pitch & 2029 Customer Story): 11 files (17.79 MB)
- `2_demonstration_part2/` (Saturday Deliverable 2: The Demonstration (Scenario Day 13.11.2026 Engine)): 107 files (1.92 MB)
- `3_commercial_and_one_page_sla/` (Saturday Deliverable 3: Commercial Model & One-Page Service Agreement): 8 files (982.60 KB)
- `4_energy_and_control_core/` (Challenge Part 1: Energy & Control Core Architecture & Audits): 48 files (7.35 MB)

Complete Cryptographic Inventory

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum
1	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/CUSTOMER_STORY_2029.md	6.13 KB	707c66a792dd8a72ac607c496f7ba2
2	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/CUSTOMER_STORY_2029.pdf	113.56 KB	37a34637b367da08667371d01db850
3	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/PITCH_DECK_OUTLINE_v2.md	44.95 KB	204e2a9c5640f609e12c9908903296
4	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/PITCH_DECK_OUTLINE_v2.pdf	667.68 KB	34cd27b347fa9a87734a29e831051c
5	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/PPT_SPEECH_PLAN.md	33.13 KB	a6ce68c132c0a78c6f99a236924df6
6	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/PPT_SPEECH_PLAN.pdf	461.51 KB	eebaf2f2515164ce81e2657ed04d78
7	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/Project_Sentinel_12pages_v3.pptx	16.06 MB	9582147dc0ea0de3d668521334c68c
8	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/Project_Sentinel_Pitch_Script_12pages.md	14.79 KB	7a684095e1e6e66e6576a33090932b
9	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/Project_Sentinel_Pitch_Script_12pages.pdf	169.58 KB	cddb4826855c89e1b23886d7995e4c
10	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/SPEECH_10MIN_ENGLISH_OFFICIAL.md	24.88 KB	164dacc7c1fa60cb006b04a51b1c1d
11	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/SPEECH_10MIN_ENGLISH_OFFICIAL.pdf	232.94 KB	e95db4592e14b5317e921c0b016308
12	2_demonstration_part2	2_The_Demonstration_Part2/DEMONSTRATION_WALKTHROUGH.md	6.61 KB	45e53483250d8a88c5b8489313fc29
13	2_demonstration_part2	2_The_Demonstration_Part2/DEMONSTRATION_WALKTHROUGH.pdf	152.15 KB	979884f995d461cde86dfdeb1f7b81
14	2_demonstration_part2	2_The_Demonstration_Part2/SPEECH_OVERHEAT_SCENE.md	9.17 KB	d42e7d94fc6c27176052d87b432aa8
15	2_demonstration_part2	2_The_Demonstration_Part2/SPEECH_OVERHEAT_SCENE.pdf	151.03 KB	a2d06a42b1cac3973c4c0fda1d4168
16	2_demonstration_part2	2_The_Demonstration_Part2/SPEECH_SIX_SCENES.md	12.60 KB	6beabed6486dd43476d4c3bcad97c0
17	2_demonstration_part2	2_The_Demonstration_Part2/SPEECH_SIX_SCENES.pdf	144.99 KB	d842df14bc3db5cf706d298ab9e8b1
18	2_demonstration_part2	2_The_Demonstration_Part2/app.py	32.36 KB	912a1c2e7c073e003ba49a57c8cf91
19	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/.streamlit/config.toml	235 B	34824b6fab69a5c93cd790e1c9180c
20	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/README.md	2.50 KB	a8adfad81f84c4e02bd6a761186189
21	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/README.pdf	96.94 KB	6d414d6725b4b713b31479886f2846
22	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/SPEECH_OVERHEAT_SCENE.md	9.17 KB	d42e7d94fc6c27176052d87b432aa8
23	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/SPEECH_OVERHEAT_SCENE.pdf	151.03 KB	f7a5329220a02c862c003c2438cbb3
24	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/SPEECH_SIX_SCENES.md	12.60 KB	6beabed6486dd43476d4c3bcad97c0
25	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/SPEECH_SIX_SCENES.pdf	144.99 KB	18793eaee8adeefec0aa475598243e
26	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/app.py	9.78 KB	7d244b2243656319f1f1cccff8b984
27	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/1_Night_Replay.py	16.84 KB	53a1fa5ef00026ec35a7af4aeebb64
28	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/2_Outage_Replay.py	15.19 KB	2af9ddec95691b1e0fbe1eedb1c8c0

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum
29	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/3_Price_Spike_Replay.py	3.16 KB	6ccbed7c0a862a8510c4b1fd8a3311
30	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/4_Curtailment_Replay.py	3.14 KB	bbb06054f14053392b36b7aebccc51
31	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/5_Cable_Fault_Replay.py	3.22 KB	af44344ba7e4cf3c06acba2b351c6c
32	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/6_Commissioning_Replay.py	3.07 KB	1c757b29c09d7cb2ae36dbc503862f
33	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/requirements.txt	491 B	89fcc60e66fbc706deeaaa9777cc63
34	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/__init__.py	90 B	de00142fb1e50bcd4418d8b2c7263a
35	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/common.py	4.82 KB	828699c80d91ea344d6ba0ad9b8c53
36	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/test_runner.py	5.79 KB	1dcf1cb66c6df0fb0eaa57333b1dae
37	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/tier1_feature_coverage.py	20.12 KB	8223a3139b255427e58ffd605b7125
38	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/tier2_boundary_corner.py	13.35 KB	19f07c20f80a3eafeaaabace23545f
39	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/tier3_cross_feature.py	6.44 KB	4542d7bc49d5805f984319d3509435
40	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/tier4_real_world_scenario.py	8.69 KB	37576eb3980aa71881606eab7b3668
41	2_demonstration_part2	2_The_Demonstration_Part2/pytest.ini	42 B	b160d82f7811555051f8966d8e0275
42	2_demonstration_part2	2_The_Demonstration_Part2/requirements.txt	488 B	31e6ed5500590b635780438f395a0d
43	2_demonstration_part2	2_The_Demonstration_Part2/run_s1.py	10.13 KB	e0fef5a26ad049e0890b3d6331705e
44	2_demonstration_part2	2_The_Demonstration_Part2/src/__init__.py	64 B	1abc659bc8548aaa23a37b1b161f89
45	2_demonstration_part2	2_The_Demonstration_Part2/src/commercial/__init__.py	253 B	92bf037063f5bc20ddfdb80cfd5865
46	2_demonstration_part2	2_The_Demonstration_Part2/src/commercial/roi_analyzer.py	10.30 KB	a3d8f7a36b1d8de4156579e8b34406
47	2_demonstration_part2	2_The_Demonstration_Part2/src/commercial/sensitivity_analyzer.py	10.88 KB	ed1f8f98fbaffe9d90c5f8d7e24ea0
48	2_demonstration_part2	2_The_Demonstration_Part2/src/commercial/settlement_clearing.py	18.15 KB	515f22d2f0ebf93ea465a86190444a
49	2_demonstration_part2	2_The_Demonstration_Part2/src/commercial/sla_calculator.py	4.41 KB	176b755b5e9395ba85653ba7bf7546
50	2_demonstration_part2	2_The_Demonstration_Part2/src/core/__init__.py	1.99 KB	3ec1b6a99e8e63783cfaae64afcc4c
51	2_demonstration_part2	2_The_Demonstration_Part2/src/core/anti_fraud.py	9.54 KB	17f392630087311f25aae723ab44f4
52	2_demonstration_part2	2_The_Demonstration_Part2/src/core/dpr_matrix.py	29.74 KB	139fffd4616829f39f49a4128ead4c
53	2_demonstration_part2	2_The_Demonstration_Part2/src/core/mpc_optimizer.py	21.21 KB	b0a4daee6238ab1d0aea01a9abc700
54	2_demonstration_part2	2_The_Demonstration_Part2/src/core/protocols.py	40.56 KB	476f7ef80ba7ea7546a50fbe7a481c
55	2_demonstration_part2	2_The_Demonstration_Part2/src/core/self_healing.py	40.85 KB	1cfd3757821f355631abbd793ff61b
56	2_demonstration_part2	2_The_Demonstration_Part2/src/core/virtual_replug.py	25.39 KB	a7fb6e1f558e6325d89d1b05c71b68
57	2_demonstration_part2	2_The_Demonstration_Part2/src/edf_packer.py	10.87 KB	2c77ed8343639b39f75386842d1191
58	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/__init__.py	265 B	879039c8c27764d2aa6b76ed9c796c
59	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/data_loader.py	4.82 KB	89cacb4f2df1397b1cb9b3088ad55d

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum
60	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/discrete_sim.py	24.13 KB	ead22a7ccb56c684923ebca0b7c69f
61	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/fault_replays.py	24.68 KB	e38becc1669acbbc35f499822b9ef2
62	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/night_replay.py	15.50 KB	473d1b8de78046cec9b1b8911e7284
63	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/outage_replay.py	9.24 KB	57e60806400e5e2d1c0033994a29b6
64	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/scenario_engine.py	16.22 KB	1c12c39430a5c7f74444f7425bae10
65	2_demonstration_part2	2_The_Demonstration_Part2/src/ui/__init__.py	84 B	81d9ba7d47beb44af1384215a904f7
66	2_demonstration_part2	2_The_Demonstration_Part2/src/ui/replay_scene.py	22.61 KB	bd6a0fde23441b7f4af13c623e98fc
67	2_demonstration_part2	2_The_Demonstration_Part2/src/utils/__init__.py	211 B	f96583f042dbc2fc7cab4bf9558fb4
68	2_demonstration_part2	2_The_Demonstration_Part2/src/utils/paths.py	1.57 KB	7d219ac723250015622e02d9134479
69	2_demonstration_part2	2_The_Demonstration_Part2/tests/analyze_scenario_day.py	3.19 KB	9b8d5c9b34b6655cd2adb199ec55b3
70	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_bess.py	554 B	6c6f5f1532f1b68f4f75b03f32ef62
71	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_d1_9.py	374 B	415358dd9fef8a0d4e93348461592a
72	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_d1_later.py	878 B	bda9828a9aee79bb65f99ca0598f97
73	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_s1_power.py	704 B	cb16ddba8d133dedf79ae9942d3770
74	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_s3_outage.py	875 B	91f87cf90aa8b92cced44c86f0f348
75	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_t01_t04.py	404 B	e3ed4d5f76d62a9bb5dc632d71d9b6
76	2_demonstration_part2	2_The_Demonstration_Part2/tests/conftest.py	157 B	f1efac1390e32ed1e95fc156828ccb
77	2_demonstration_part2	2_The_Demonstration_Part2/tests/edf_simulation.py	3.39 KB	6b1952c1a6548386e8446a7bb7e35b
78	2_demonstration_part2	2_The_Demonstration_Part2/tests/inspect_cols.py	454 B	8be91976946d4e3a19421fe3f516d2
79	2_demonstration_part2	2_The_Demonstration_Part2/tests/inspect_dataset.py	955 B	4d3442437c17b4eb003a322ca67f54
80	2_demonstration_part2	2_The_Demonstration_Part2/tests/inspect_s1_telemetry.py	1.10 KB	f06cdaabdd4f8db9305f2b164e6950
81	2_demonstration_part2	2_The_Demonstration_Part2/tests/query_scenario_data.py	1.53 KB	a5c848437b5e132665655201603f6e
82	2_demonstration_part2	2_The_Demonstration_Part2/tests/simulate_s2_curtailment.py	2.75 KB	b88a98fb37592964b92d3802faa6ee
83	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_adversarial_discrete_sim_thermal.py	24.76 KB	123976b1868da5328a43f6cbfbe09d
84	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_adversarial_m1.py	19.24 KB	3d22b03e00023d1c208f3dd876876b
85	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_anti_fraud_replug.py	5.19 KB	93b485cc6dfc88184f95686be1c720
86	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_commercial_settlement.py	15.60 KB	4c73d7bdd9fec16867f1151098a993
87	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_dashboard_runtime_stress.py	14.51 KB	ce3f40c9a5c25b6d60c1baff3dd868
88	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_demo_day_simulation.py	19.28 KB	bbb4afbd86d92b222b1ef4cd8ed58a
89	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_discrete_sim_engine.py	8.00 KB	c6012da71b0f8dc933557e9447236a
90	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_dpr_stress_adversarial.py	19.89 KB	666effd97188e1f1534e94ff1fd7e1
91	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_e2e_suite.py	45.75 KB	10b7ad1b2b5c69f94687a08ac158bf

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum
92	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_gate2_empirical_verification.py	6.97 KB	c4a1e8f11cc9101874febb8a21b534
93	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_github_issue2_29items.py	14.33 KB	243ef14cb284c97473a0a12f0ce3a5
94	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m1_adversarial_probe.py	21.96 KB	879b0152b52209f98b372a0ea5860b
95	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m1_remediation.py	13.75 KB	c408e705ba07c7f3f8aa9998e9bb73
96	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m2_challenger_empirical.py	21.39 KB	3daf44cf881c14f7532e2966fdd5e0
97	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m2_hardware_edge_stress.py	17.97 KB	f7118be5fb53d1be6ab324d9b968d7
98	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m2_hardware_self_healing.py	16.38 KB	ab2808b81adaad14d89ae56bd5c73e
99	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m2_protocols_anti_fraud_adversarial.py	19.76 KB	0370eeef177b3a24cd67b85513649d
100	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m3_adversarial_stress.py	16.52 KB	218df08d523ca666d4437d6e0351bd
101	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_mpc_math_fixes.py	5.64 KB	a721eaf0bb90da158a2e37d91ae2c3
102	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_protocols_ocpp_iso.py	14.93 KB	dbdddd0672d752b41e1f95372a6828
103	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_remediated_sla.py	5.71 KB	b304876d5bd0d4742c288319e51963
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105	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_tier5_commercial_adversarial.py	42.05 KB	c3f3871bf19d77315e2a5ab9d9dc33
106	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_tier5_core_adversarial.py	43.93 KB	a0838458f9ae65f9184428aea32a84
107	2_demonstration_part2	2_The_Demonstration_Part2/tests/verify_csv_inventory.py	2.94 KB	4c97df1cf3dd36ae7cb8aac3c9ea9b
108	2_demonstration_part2	2_The_Demonstration_Part2/tests/verify_empirical_claims.py	7.84 KB	c22ca88a0dd01ca42b8601e615e75a
109	2_demonstration_part2	2_The_Demonstration_Part2/tools/check_demo_artifacts.py	3.12 KB	7f35cdc91661faece79eaf71b2536d
110	2_demonstration_part2	2_The_Demonstration_Part2/tools/cjk_audit.py	1.41 KB	3ba5b8cd1979b16a15df80afa705f1
111	2_demonstration_part2	2_The_Demonstration_Part2/tools/export_deck_ledger.py	39.59 KB	a20260946a57c8857add3b5e3d5b70
112	2_demonstration_part2	2_The_Demonstration_Part2/tools/generate_one_page_docx.py	9.72 KB	7523092e099ad3555eb3385e39071b
113	2_demonstration_part2	2_The_Demonstration_Part2/tools/md_to_pdf.py	5.74 KB	02a2f804faa6655534301dcf6639ba
114	2_demonstration_part2	2_The_Demonstration_Part2/tools/package_and_manifest.py	16.72 KB	e9378f3f62e06053ea2e5151362afa
115	2_demonstration_part2	2_The_Demonstration_Part2/tools/s1_multicabinet_sim.py	12.12 KB	ea9c9c403c849dd14cecfb3ea9435c
116	2_demonstration_part2	2_The_Demonstration_Part2/tools/stage_submission.py	8.10 KB	42887b990fb02a8092f88eef9791cd
117	2_demonstration_part2	2_The_Demonstration_Part2/tools/verify_submission_cjk.py	2.33 KB	b142ee1ac6b2c129459ca444d7e793
118	2_demonstration_part2	2_The_Demonstration_Part2/verify_all.py	6.76 KB	f8d05f4dd439d48773ad4aafdd4346
119	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/COMMERCIAL_BUSINESS_MODEL.md	4.97 KB	61d6db3b85b684af95aee906ffaa23

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum
120	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/COMMERCIAL_BUSINESS_MODEL.pdf	116.12 KB	11310f8f3f93b6e7c1e63aab4ee6b2
121	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/One_Page_SLA_Specification.md	22.20 KB	65fe05c7e3269eea74359e9f966929
122	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/One_Page_SLA_Specification.pdf	634.92 KB	3aec70437c58090038e975bf22634b
123	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/Team3-One_Page-Autonomous_Charging_Services_Agreement.docx	38.17 KB	b20b09a2fae74d3ba8799efb1bb2d4
124	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/one_page_agreement.html	11.08 KB	a92cb6b5c7427ea02d216cd5d97ea4
125	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/one_page_agreement.md	6.58 KB	254e3961995d99d9d151b2db2f824a
126	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/one_page_agreement.pdf	148.54 KB	95eed0f121af5cec93d8a79fa224c8
127	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/HACKATHON.md	3.72 KB	be16e96cf571ffabb8f728d489dc26
128	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/HACKATHON.pdf	152.82 KB	7e4a4bffbfc04333cc9ec1804ceca0
129	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/PROJECT.md	17.15 KB	1804e340c355504f911c24279dc490
130	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/PROJECT.pdf	224.51 KB	090beb32e3ac4ef433449771dbbdc8
131	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/PROJECT_AUDIT_REPORT.md	7.17 KB	9ee879203249404d8ac5b1c49ea2be
132	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/PROJECT_AUDIT_REPORT.pdf	142.42 KB	c874ee89f4d5fa3fee50d028085523
133	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/README_EN.md	35.02 KB	1847b44b5456b23b5bed36f95b12c5
134	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/README_EN.pdf	604.49 KB	641617a5779928185dcc7952e6a4ca
135	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/ZERO_HUMAN_DESIGN.md	18.70 KB	4a1df3da519f8b69c763474128e7fc
136	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/ZERO_HUMAN_DESIGN.pdf	312.53 KB	624c91acfd31cd3a84daaee831dbe9
137	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/assumptions_and_additions.md	11.67 KB	b51226c0eabfe14f722e204828169e
138	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/assumptions_and_additions.pdf	277.50 KB	15011c95aa96a7a08a44a827452e23
139	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/attribution_matrix.md	13.94 KB	c589f83d68fa7cfcada147a9c1c7e4
140	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/attribution_matrix.pdf	466.86 KB	28e2bcf75133c901623a241d2514f8
141	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/00_pocket_cheat_sheet.md	8.88 KB	8c1e41f37a71acf3d5591d7c2eaed0
142	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/00_pocket_cheat_sheet.pdf	157.29 KB	b80f5a55acaebe8c52ca74e83d8ea0
143	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/01_stage_schedule_and_setup.md	47.69 KB	94bd789edce7e64106fb7bb345c0c8
144	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/01_stage_schedule_and_setup.pdf	433.11 KB	ba9d87b467b95788b5cd11f8d240d4
145	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/02_team_roles_and_coordination.md	32.92 KB	35a0133c5902c568cfdc30d3084817
146	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/02_team_roles_and_coordination.pdf	367.02 KB	76418eee8c8f294af5e8dc184f4072
147	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/03_judge_personas_and_question_bank.md	31.10 KB	fbe274474571ad072923fd77e3f4a0

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum
148	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/03_judge_personas_and_question_bank.pdf	305.30 KB	d207de84f17dbe8f1d00de5332c97e
149	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/04_structured_defense_and_crisis_manual.md	48.96 KB	bfde4d27f643c05125d413f31e3906
150	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/04_structured_defense_and_crisis_manual.pdf	571.60 KB	0fd7488dc9e11be67b2f9b429bd5fe
151	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/05_live_demo_operational_sop.md	12.56 KB	d80535b094e780b7ba1b0c5dfe0cf0
152	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/05_live_demo_operational_sop.pdf	288.99 KB	b73b9b7c70cdab5e49a2e63cfceb05
153	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/06_code_proof_and_authenticity_guide.md	12.18 KB	1aafcb5d859dae4de823a21f10d3a0
154	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/06_code_proof_and_authenticity_guide.pdf	403.17 KB	f083312c3b9ddcde07fb6bb501de00
155	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/PROJECT_SENTINEL_MASTER_PORTRAIT_AND_FILE_INDEX_EN.md	35.02 KB	1847b44b5456b23b5bed36f95b12c5
156	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/PROJECT_SENTINEL_MASTER_PORTRAIT_AND_FILE_INDEX_EN.pdf	604.49 KB	6079ede685a8fbb7fc6726f3a22838
157	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/README.md	16.53 KB	14e87f49a9ca2ea5871ea0a355b675
158	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/README.pdf	180.53 KB	50c6cb5c462423e5561b04743b08e3
159	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/energy_impact.md	20.66 KB	deffb9be6a910a30dc7f277e1bca72
160	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/energy_impact.pdf	536.04 KB	7665a53f3e2ae5c06dcb1c4e2733c8
161	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/BRIEF.md	6.18 KB	571c27eb8738cb0b1a6d1ce3775c6a
162	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/BRIEF.pdf	193.66 KB	f7c60221d65042f16bf900925dc05a
163	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/BUILD_PLAN.md	2.31 KB	11af160f1836588db876ec8d590c70
164	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/BUILD_PLAN.pdf	115.57 KB	1dd1b0aab0daf7fcffc1b38824bbc1
165	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/DECISIONS.md	3.41 KB	db08a4e7f78a9724bc30a86188fabc
166	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/DECISIONS.pdf	156.95 KB	554bbbab1e0c4fae1b9c1e94a5afb9
167	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/DEMO_SCRIPT.md	5.61 KB	4b3979ec02a20e050d04812cc9ab0f
168	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/DEMO_SCRIPT.pdf	138.88 KB	6f5e5b4a222c981ad735280c7414ab
169	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/JUDGING.md	5.51 KB	c6b827e9d0665a249fc563c9e00e0c
170	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/JUDGING.pdf	173.19 KB	607dc377f711eace15965db94699bd
171	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/RULES.md	3.51 KB	76170f3d29259f6c0232486e31fbc9
172	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/RULES.pdf	135.55 KB	91cfe99e7d8771911872651b030b2f
173	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/SUBMISSION.md	5.32 KB	13f6abe5fee18c06ecbc981db851a4
174	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/SUBMISSION.pdf	177.94 KB	8cc46dc94e9ae1ef1a3e003746d70f
175	root	README.md	24.05 KB	d1433f2282812746cfeb884e428e96

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum
176	`root`	`README.pdf`	583.62 KB	`59fbbda8adb558f38ad52708fd639a`

Independent Verification Instructions

To independently verify the integrity of all files in this submission, run:

```
# 1. Unzip the official archive
unzip Team3_Project_Sentinel_Submission.zip -d submission_verify/
cd submission_verify/2_The_Demonstration_Part2/

# 2. Run the deterministic verifier (< 0.5s)
python verify_all.py

# 3. Run the complete E2E and Pytest test suites
python -m e2e_tests.test_runner
pytest tests/ -q
```